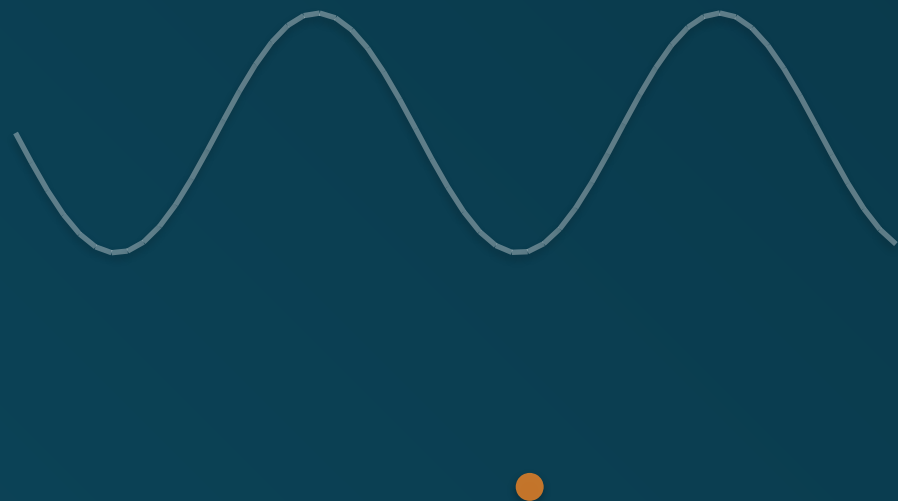


TOPIC 24

Medical Physics

医学物理



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

24.1 Ultrasound in medicine 医学超声

- Piezo-electric effect: generating and detecting ultrasound 压电效应
- Specific acoustic impedance $Z = \rho c$ and intensity reflection coefficient 声阻抗与反射系数
- Attenuation $I = I_0 e^{(-\mu x)}$; A-scan and B-scan 衰减与成像方式

24.2 X-rays in medicine 医学 X 射线

- Production of X-rays; minimum wavelength $\lambda_{\min} = hc/eV$ X 射线产生
- Attenuation mechanisms; improving image contrast 衰减与影像对比
- CT scanning: image reconstruction CT 扫描

24.3 PET scanning PET 扫描

- Radioactive tracers; positron emission and annihilation 示踪剂与湮灭
- Detection of the two 511 keV γ -ray photons 符合探测

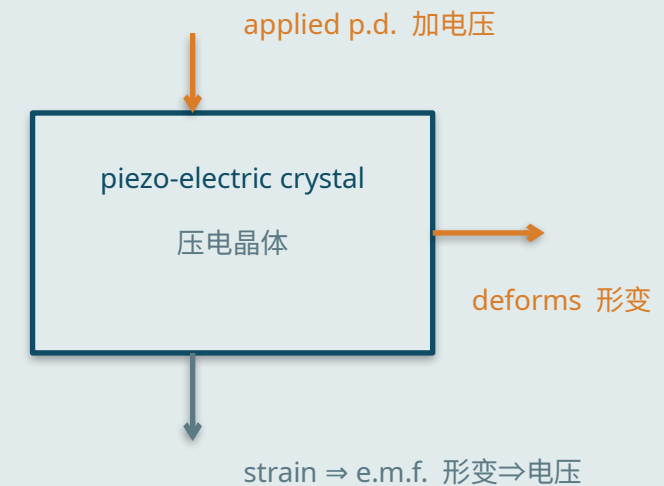
The Piezo-electric Effect

压电效应

A **piezo-electric crystal** (e.g. quartz, PZT) changes shape when a p.d. is applied across it, and produces an e.m.f. across its faces when it is compressed or stretched.

压电晶体：加电压 $\Rightarrow$ 形变；受形变 $\Rightarrow$ 产生电动势，效应可逆。

- The effect is **reversible** — the same crystal generates and detects ultrasound 收发两用
- Applying an alternating p.d. makes the crystal vibrate 交变电压使晶体振动
- Vibration amplitude is greatest at the crystal's **resonant frequency** 共振时振幅最大



Reversible piezo-electric effect 可逆压电效应

Generating & Detecting Ultrasound

超声的产生与接收

An ultrasound **transducer** uses a piezo-electric crystal. An alternating p.d. at the crystal's resonant frequency produces an ultrasound pulse; a returning echo makes the crystal vibrate and generates an electrical signal.

换能器：交变电压⇒超声脉冲；回波⇒电信号，同一晶体收发分时工作。

- Ultrasound = sound waves with frequency above 20 kHz; medical imaging uses 1–15 MHz 医学超声频率
- The crystal is damped to produce **short pulses** — transmission and reception share the same crystal 短脉冲
- Echo delay time ⇒ distance to the reflecting boundary: $\text{depth} = \frac{1}{2}ct$ 回波延时⇒深度
- Pulse repetition frequency limited by echo return time 脉冲重复频率受限

Ultrasound Imaging

超声成像原理

- Pulses reflect at boundaries between different tissues 组织界面反射
- Echo delay $\Rightarrow$ depth; echo amplitude $\Rightarrow$ boundary type 延时定深度、幅度辨界面
- **A-scan**: 1-D graph of echo amplitude vs time 一维幅度显示
- **B-scan**: 2-D image built from many A-scans as the probe moves 二维图像
- Advantages: no ionising radiation, real-time, relatively cheap 无电离辐射



Specific Acoustic Impedance

声阻抗与反射

Specific acoustic impedance Z of a medium is the product of its density ρ and the speed of sound c in it (unit $\text{kg m}^{-2} \text{s}^{-1}$).

声阻抗 $Z = \rho c$ ，单位 $\text{kg m}^{-2} \text{s}^{-1}$ 。

$$\frac{I_R}{I_O} = \dots$$

垂直入射时的强度反射系数

- Large mismatch (air ↔ tissue) ⇒ almost total reflection 空气界面几乎全反射
- Coupling gel excludes air so ultrasound can enter the body 耦合剂排除空气
- Bone-muscle boundary: $\alpha \approx 0.34$ — enough reflection for imaging 骨界面反射适中
- Typical $Z / 10^6 \text{ kg m}^{-2} \text{s}^{-1}$: air ≈ 0.0004 , fat 1.4, muscle 1.7, bone ≈ 6.5 典型数值

Attenuation of Ultrasound

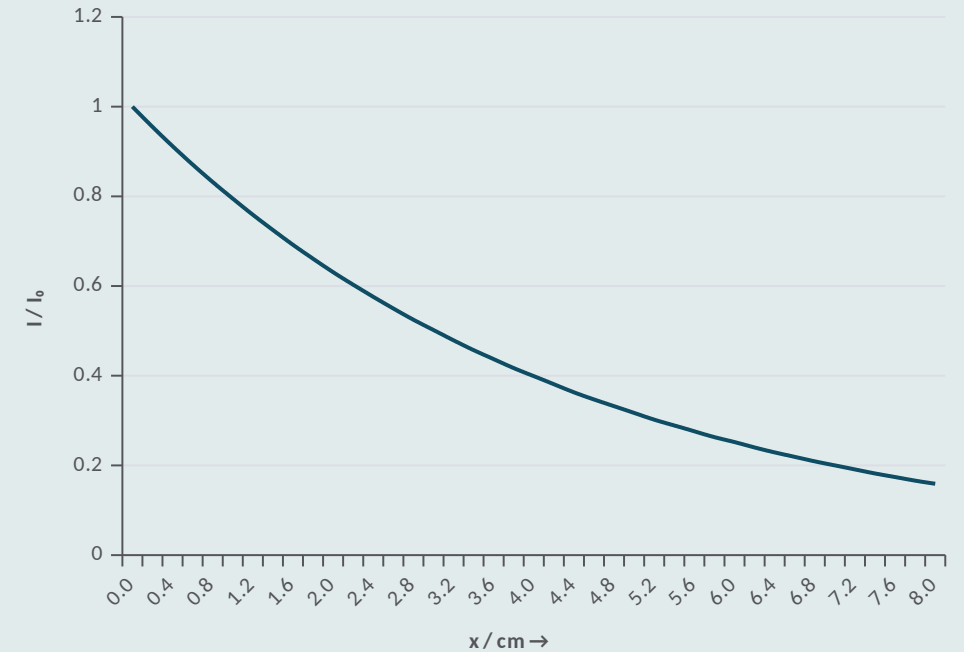
超声的衰减

As a parallel beam passes through tissue, absorption reduces its intensity **exponentially**. μ is the linear absorption (attenuation) coefficient of the medium.
平行束强度随深度指数衰减。

$$I = I_0 e^{-\mu x}$$

μ 取决于介质与频率

- Higher frequency $\Rightarrow$ better resolution but larger μ (less penetration) 频率：分辨率 $\leftrightarrow$ 穿透深度
- Typical μ (muscle) $\approx 0.23 \text{ cm}^{-1}$ at 1 MHz 典型数值
- Formula applies to a parallel beam; divergent beams also lose intensity by spreading 平行束条件



Exponential attenuation in tissue 组织中的指数衰减

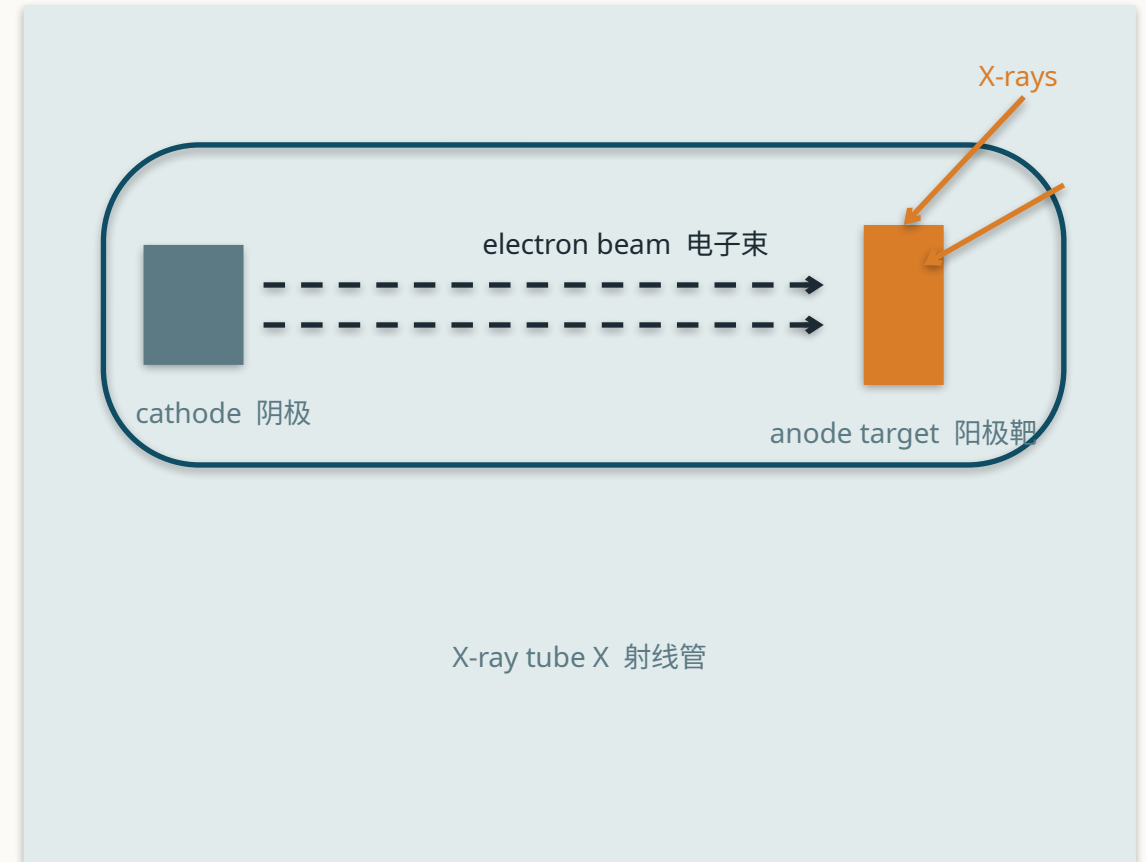
Production of X-rays

X 射线的产生

X-rays are produced when **high-speed electrons are suddenly decelerated** by hitting a metal target (anode) in an evacuated X-ray tube.

高速电子轰击金属靶、急剧减速时产生 X 射线。

- Hot cathode emits electrons (thermionic emission) 热阴极发射电子
- High p.d. (30–150 kV) accelerates electrons to the anode 高压加速
- Only $\approx 1\%$ of electron energy becomes X-rays — the anode needs cooling 效率低需冷却
- Rotating anode spreads the heat 旋转阳极散热

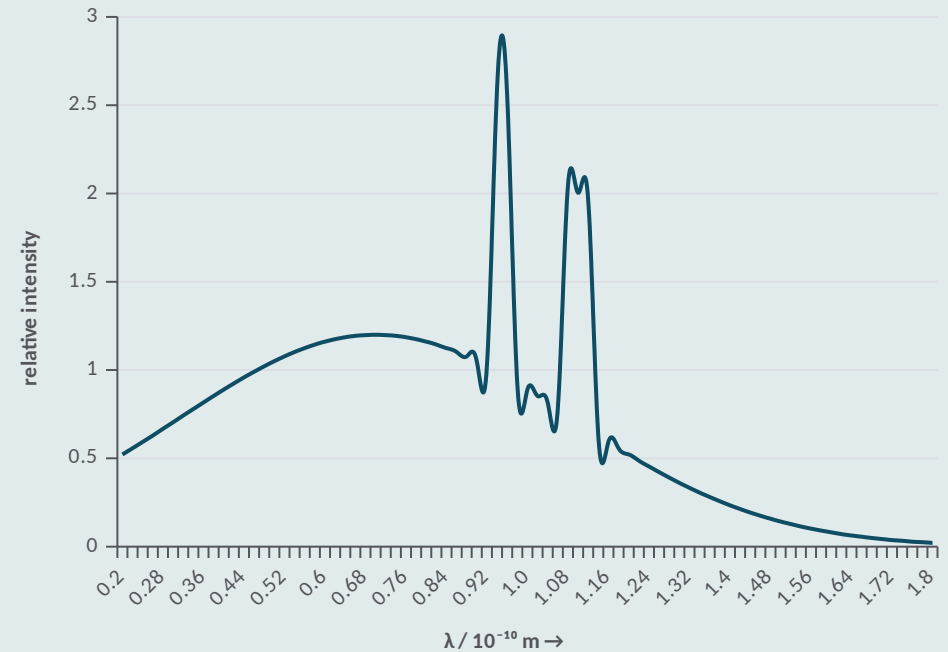


The X-ray Spectrum

X 射线谱

- **Continuous spectrum:** bremsstrahlung — electrons decelerated in the target 连续谱（轫致辐射）
- **Characteristic lines:** target atoms' inner-shell transitions 特征谱线
- Sharp cut-off at the **shortest wavelength** 存在最短波长
- $\uparrow$ tube voltage $\Rightarrow$ shorter $\lambda_{\min}$ (harder X-rays); $\uparrow$ current $\Rightarrow$ higher intensity 管压定硬度

$$\lambda_{\min} = \frac{hc}{eV} \quad \text{全部电子动能转为一个光子}$$



Continuous spectrum + characteristic peaks 连续谱 + 特征峰

Attenuation of X-rays

X 射线的衰减与成像

$$I = I_0 e^{-\mu x}$$

与超声衰减同形式

- μ depends on the medium and on photon energy μ 与介质及光子能量有关
- Bone attenuates much more than soft tissue $\Rightarrow$ shadow on film 骨吸收强 $\Rightarrow$ 影像对比
- X-ray image = 2-D **shadow**, not a lens image 影像是 " 影子 "
- Aluminium filter removes long-wavelength photons that only add dose 铝滤片去软射线
- Contrast improved by contrast media (e.g. barium meal) 造影剂增强对比

Computed Tomography (CT)

CT 扫描

In **CT scanning**, X-ray images of a thin slice of the body are taken from many different angles, and a computer reconstructs a detailed cross-sectional image.

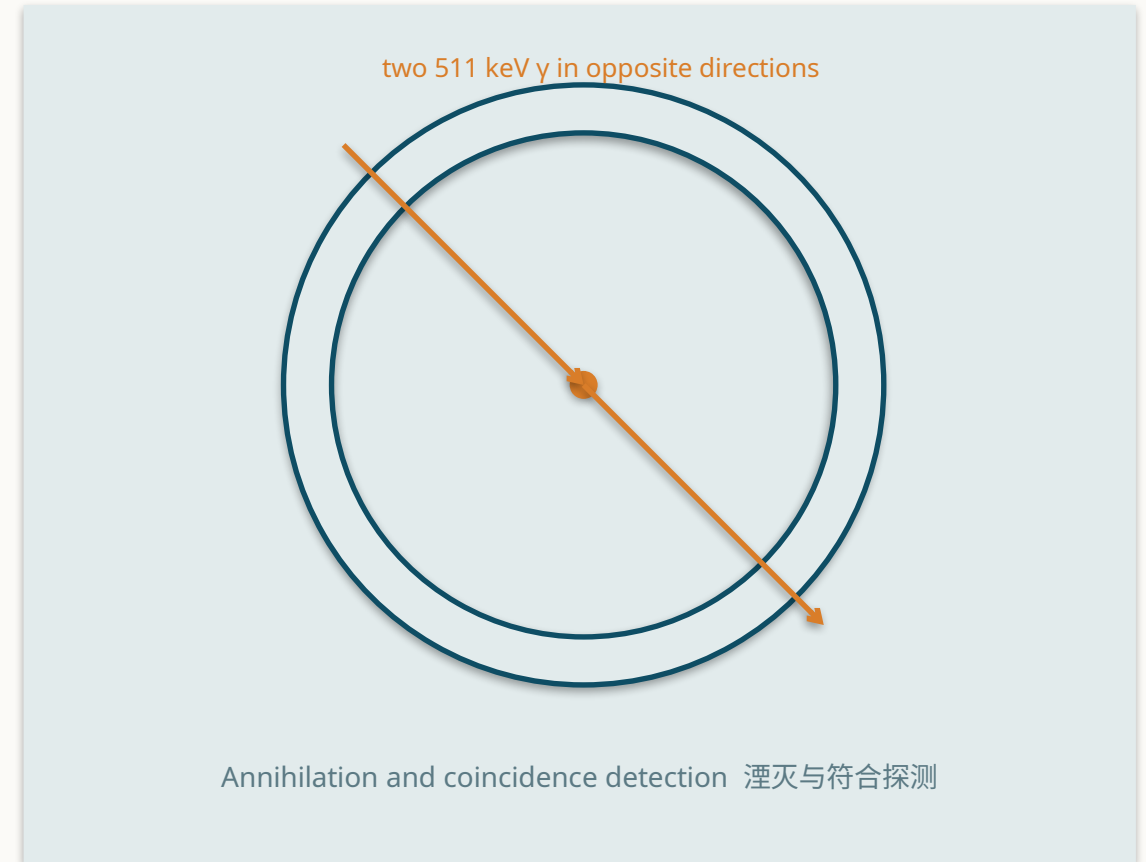
多角度投影 + 计算机重建断层图像。

- The slice is divided into **voxels**; each voxel's attenuation is computed 体素重建
- Millions of calculations $\Rightarrow$ powerful computer required 计算量大
- Advantage over plain X-ray: distinguishes **soft tissues** of similar density 软组织分辨
- 3-D image can be viewed from any angle; brightness/contrast adjustable 三维任意视角
- Disadvantage: larger radiation dose than a single X-ray 剂量较大

PET Scanning

PET 扫描原理

1. Tracer with a **β^+ emitter** (e.g. ^{18}F -FDG) is injected and collects at active tissue 正电子示踪剂
 2. Emitted **positron annihilates** with an electron 正电子与电子湮灭
 3. Annihilation produces **two 511 keV γ -photons travelling in opposite directions** 两个反向 511 keV 光子
 4. Ring detectors register photons **in coincidence**; the source lies on the line joining the detectors 符合探测定位
- PET maps **metabolic activity**, not anatomy 代谢功能成像



Worked Example 24A

例题解析

WORKED EXAMPLE 24A

Using $Z(\text{air}) = 430 \text{ kg m}^{-2} \text{ s}^{-1}$, $Z(\text{soft tissue}) = 1.6 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$, calculate the intensity reflection coefficient for a parallel beam incident normally on: a) the air–soft tissue boundary; b) a muscle–bone boundary ($Z(\text{muscle}) = 1.7 \times 10^6$, $Z(\text{bone}) = 6.5 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$).

这就是检查时必须涂耦合剂排除空气的原因。

SOLUTION 解答

Step 1

Step 2 a) $\alpha = (1.6 \times 10^6 - 430)^2 / (1.6 \times 10^6 + 430)^2 \approx 0.999 \approx 1.0$ — almost total reflection

Step 3 b) $\alpha = (6.5 - 1.7)^2 / (6.5 + 1.7)^2 = (4.8/8.2)^2 = 0.34$

Answer a) ≈ 1.0 ; b) 0.34

Worked Examples 24B & 24C

例题解析

WORKED EXAMPLE 24B

Ultrasound passes through 3.5 cm of muscle, reflects at a bone surface ($\alpha = 0.34$) and returns through the muscle. For muscle $\mu = 0.23 \text{ cm}^{-1}$, find the fraction of the incident intensity that arrives back.

1. Total path = 7.0 cm: $I/I_0 = e^{(-0.23 \times 7.0)} = 0.20$
2. Reflected fraction at bone = 0.34
3. Returning fraction = 0.20×0.34

Answer $I/I_0 = 0.068$

WORKED EXAMPLE 24C

The accelerating p.d. of an X-ray tube is 30 kV. Calculate the minimum wavelength of photons in the beam. ($h = 6.63 \times 10^{-34} \text{ J s}$, $e = 1.60 \times 10^{-19} \text{ C}$, $c = 3.00 \times 10^8 \text{ m s}^{-1}$)

1. Maximum photon energy = electron kinetic energy: $eV = hc/\lambda_{\min}$
2. $\lambda_{\min} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (1.60 \times 10^{-19} \times 30 \times 10^3)$

Answer $\lambda_{\min} = 4.1 \times 10^{-11} \text{ m}$

Worked Examples 24D & 24E

例题解析

WORKED EXAMPLE 24D

The linear absorption coefficient of copper is 0.693 mm^{-1} . Calculate: a) the thickness that reduces intensity by 50%; b) the fraction transmitted through 1.2 cm of copper.

1. a) $0.50 = e^{(-0.693x)} \Rightarrow x = \ln 2 / 0.693 = 1.0 \text{ mm}$
2. b) $I/I_0 = e^{(-0.693 \times 12)} = e^{(-8.3)}$

Answer a) 1.0 mm; b) 2.4×10^{-4}

WORKED EXAMPLE 24E

Compare the image produced during an X-ray investigation with that produced in CT scanning.

1. X-ray: 2-D projection (shadow) of a 3-D object onto a flat detector
2. CT: computer-reconstructed 3-D image; sections can be viewed from many angles, with adjustable contrast

Answer 2-D shadow vs 3-D reconstruction

Class Practice

随堂练习

- Q1** Calculate the intensity reflection coefficient for ultrasound incident normally on a fat-muscle boundary ($Z_{\text{fat}} = 1.4 \times 10^6$, $Z_{\text{muscle}} = 1.7 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$).
- Q2** Water has $\mu = 0.29 \text{ cm}^{-1}$ for certain X-ray photons. Calculate the depth of water that reduces the intensity of a parallel beam to 3.0×10^{-3} of its incident value.
- Q3** Put the stages of PET imaging in order and explain the role of coincidence detection: positron emission / two 511 keV photons / injection of tracer / line of response located / annihilation.

Answers Q1: $(0.3/3.1)^2 = 9.4 \times 10^{-3}$ Q2: $x = \ln(333)/0.29 = 20 \text{ cm}$ Q3: 注射→正电子→湮灭→双光子→符合探测定响应线

Summary

本章小结

KEY FORMULAS 核心公式

$$Z = \rho c$$

声阻抗

$$\frac{I_R}{I_0} = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

强度反射系数

$$I = I_0 e^{-\mu x}$$

指数衰减

$$\lambda_{min} = hc / eV$$

X 射线最短波长

$$E_\gamma = 511 \text{ keV}$$

湮灭光子能量

$$depth = \frac{1}{2} ct$$

回波测深

KEY IDEAS 核心思想

- Piezo-electric crystal both generates and detects ultrasound 压电晶体收发一体
- Coupling gel removes the air gap ($\alpha \approx 1$ at air boundary) 耦合剂排除空气
- X-ray = 2-D shadow; CT = 3-D reconstruction; PET = metabolic map 三种影像对比
- Two opposite 511 keV photons are the key to PET localisation PET 定位关键

Exam Tips

备考提示

- Reflection coefficient: square the **difference** first, then divide by the square of the **sum** — order matters. 先差后和的平方。
- For $I = I_0 e^{-\mu x}$: check that μ and x have consistent units (cm^{-1} with cm). μ 与 x 单位匹配。
- Ultrasound uses pulses, not continuous waves, so that echoes can return before the next pulse is sent. 脉冲而非连续波。
- PET: the two photons are **simultaneous and opposite** — both facts are needed for locating the source. 同时 + 反向缺一不可。